

PHANEENDRA GANJI

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI Engineer with 4+ years of experience designing and deploying production-scale AI, machine learning, and data science solutions across fintech and healthcare domains. Expertise in Generative AI, Agentic AI, LLMs, NLP, predictive modeling, MLOps, and cloud-native ML systems using Python, AWS, Kubernetes, and modern ML frameworks. Proven track record of building scalable data platforms and intelligent automation solutions that improve operational efficiency, enhance customer experience, and deliver measurable business impact. Strong collaborator with cross-functional product, engineering, and domain teams, experienced in taking AI products from research and development through production deployment while ensuring responsible AI governance and regulatory compliance.

PROFESSIONAL EXPERIENCE

AI Engineer | Intuit | Bellevue, WA | July 2024 – Present

- Architected the end-to-end agentic AI pipeline for QuickBooks automated invoice processing by designing a multi-agent orchestration layer atop GenOS, enabling autonomous extraction, validation, and reconciliation of accounts payable data across 2M+ small business customer accounts.
- Engineered classical gradient boosting and transformer-based document understanding models to extract invoice line items from unstructured PDFs and emails, achieving 94.5% field-level accuracy and reducing manual data entry errors by 35% for early beta users.
- Developed an LLM-based reasoning agent using Anthropic Claude via AWS Bedrock to resolve ambiguous payment discrepancies and automate exception handling, decreasing accounts receivable resolution time by 28% and improving customer confidence scores by 18%.
- Built scalable real-time inference microservices in Python and Go deployed on AWS EKS with auto-scaling, reducing average model serving latency from 850ms to 210ms while supporting 450K+ daily predictions during peak tax-season traffic.
- Implemented comprehensive guardrails and model governance frameworks including bias detection across business demographics and automated model cards, ensuring compliance with Intuit's Responsible AI principles and passing Model Review Board audits on first submission.
- Partnered with product managers, tax domain experts, and QuickBooks UX designers to integrate agentic workflows into the existing reconciliation UI, driving 42% month-over-month adoption growth and reducing customer support tickets related to invoicing by 31%.
- Led the production rollout through canary deployments and A/B experimentation, measuring a 22% reduction in days-sales-outstanding and \$18M annualized time savings for small business customers, while maintaining 99.97% system uptime across distributed AWS regions.

Senior Data Scientist | Optima Financial Services, India | Sep 2021 – July 2022

- Built a centralized client analytics platform by integrating KYC, financial goal, and transaction data from Salesforce CRM and Tally ERP into a PostgreSQL data warehouse using Python-based ETL pipelines, reducing data aggregation time by 40% and eliminating manual Excel-based reporting for 15 financial advisors.
- Developed an XGBoost-based client risk profiling engine by engineering behavioral and demographic features from 8,000+ onboarding records, validated against SEBI suitability guidelines, achieving 86% classification accuracy and reducing manual risk assessment time per client from 45 minutes to 12 minutes.
- Built a goal-based mutual fund recommendation system by combining client risk scores with time-horizon analysis and fund performance metrics, deploying via REST APIs integrated into the advisor portal, resulting in a 32% increase in product cross-sell rate and a 22% improvement in client portfolio diversification scores.
- Established automated model monitoring and quarterly retraining pipelines using MLflow and cron-scheduled Python scripts, implementing drift detection and explainability reports for SEBI audit trails, maintaining model F1 scores within 2% of baseline and ensuring zero compliance violations during regulatory reviews.

Data Scientist | Medvora Healthcare, India | Jan 2020 – Aug 2021

- Implemented centralized ETL pipelines using Python and Apache Airflow to unify patient demographics, appointment logs, and insurance claims from 8 disparate clinical systems into AWS S3 and PostgreSQL, reducing data silos by 60% and enabling real-time analytics for 120,000+ patient records.
- Developed XGBoost ensemble models to predict patient no-shows and wellness program dropout risk with 84% AUC-ROC, enabling care coordinators to implement targeted outreach that reduced missed appointments by 31% and improved weight management program retention by 24%.

- Engineered a clinical NLP pipeline using scikit-learn and regex-based entity extraction to parse 45,000+ unstructured SOAP notes across primary care and dental units, collaborating with physicians to validate automated symptom tagging and reducing manual documentation review time by 38%.
- Designed K-means clustering algorithms to segment patients by chronic disease risk and insurance utilization patterns, partnering with care management teams to deploy personalized outreach strategies that increased preventive screening compliance by 27% and reduced per-patient service costs by 18%.
- Productionized machine learning models via Python Flask APIs deployed on AWS EC2 with automated monitoring dashboards in Tableau, enabling real-time risk scoring for 12,000+ daily patient interactions while maintaining strict PHI de-identification protocols and role-based access controls.

TECHNICAL SKILLS

Core AI & GenAI: Agentic Workflows & Multi-Agent Orchestration, Large Language Models (LLMs), Generative AI (GenAI), Retrieval-Augmented Generation (RAG), Fine-Tuning & Prompt Engineering, Document Understanding, Natural Language Processing (NLP), Computer Vision, Classical Machine Learning (Gradient Boosting / XGBoost, K-Means Clustering), Risk Profiling & Scoring Systems.

ML Frameworks & Toolkits: PyTorch, Hugging Face Transformers, Scikit-learn, XGBoost, MLflow, LangChain/LlamaIndex, AWS Bedrock (Anthropic Claude).

Systems & Infrastructure: Microservices Architecture, Model Serving & Latency Optimization, Real-Time Inference, Distributed Systems, Containerization, Auto-scaling, Model Monitoring, Drift Detection.

Data Engineering & Pipelines: Orchestration & ETL, Pipeline Automation, Feature Engineering, Unstructured Data Parsing, Data Warehousing, Data De-identification & Compliance.

Languages & Querying: Python, Go, SQL, Bash.

Cloud, DevOps & Data Platforms: AWS (EKS, EC2, S3, Bedrock), PostgreSQL, Apache Airflow, Docker, Kubernetes, Tableau, Git/GitHub, REST APIs.

MLOps, Governance & Compliance: Responsible AI, Model Cards, Bias & Fairness Detection, Model Governance, Audit Trail Generation, Regulatory Compliance (SEBI, HIPAA/PHI).

PROJECTS

Production RAG Knowledge Assistant | LangChain · Pinecone · GPT-4o · FastAPI · Streamlit · Docker

- Built an end-to-end **enterprise RAG system** ingesting 100K+ documents into a **Pinecone vector index** (text-embedding-3-large); served via a **FastAPI backend** with sub-300ms p95 retrieval latency and a Streamlit chat UI
- Implemented **hybrid dense-sparse search** (BM25 + vector similarity), improving retrieval precision by 22% over pure vector search; integrated **Langfuse for LLM observability** and hallucination tracking across 10K+ daily queries

LLM Fine-Tuning Pipeline — Domain-Specific QA Bot | LoRA / QLoRA · Mistral-7B · MLflow · Hugging Face PEFT · W&B

- Fine-tuned **Mistral-7B using QLoRA (4-bit quantization + LoRA adapters)** on a 15K-sample domain Q&A dataset; achieved **94.2% answer accuracy** vs. 78% zero-shot baseline while reducing GPU memory footprint by 65%
- Tracked all experiments with **MLflow** and **Weights & Biases**; deployed quantized model via **vLLM inference server** on AWS EC2, achieving 3× throughput vs. standard Hugging Face pipeline

Multi-Agent Agentic AI System | LangGraph · CrewAI · OpenAI API · Tavily Search · FastAPI

- Architected a **multi-agent orchestration system using LangGraph** with specialized agents for research, analysis, and code generation; automated a 5-step report workflow saving **20+ analyst hours per week**
- Integrated **tool-calling APIs** (Tavily web search, custom Python REPL, PDF parser) with state-graph memory management; system handled 500+ concurrent sessions with consistent response quality

CERTIFICATIONS

Generative AI with Large Language Models – DeepLearning.AI / Coursera | [Certificate](#)

Complete MLOps Bootcamp – Udemy / Zero to Mastery | [Certificate](#)

Neural Networks and Deep Learning – DeepLearning.AI | [Certificate](#)

Databricks Certified Data Engineer Associate – Databricks | [Certificate](#)

AWS Fundamentals Specialization – Amazon Web Services | [Certificate](#)

Databricks Data Engineering | [Certificate](#)

Python Programming Professional Certificate – Coursera | [Certificate](#)

EDUCATION

Master of Science in Data Science and Analytics | Florida Atlantic University

May 2024

Relevant coursework: Machine Learning, Deep Learning, NLP, Big Data Analytics, Statistical Modeling

Post Graduate Diploma in ML & AI | IIIT Bangalore

Aug 2022